

Akaike and Bayesian Information Criteria: A Rigorous Elementary Derivation*

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1 Introduction and Motivation

Model selection is a central problem in statistics: given data $\mathbf{y} = (y_1, \dots, y_n)$ and a collection of candidate parametric models $\{M_1, M_2, \dots\}$, which model should we prefer?

A naive answer is to maximise the log-likelihood $\ell(\hat{\theta}; \mathbf{y})$ over the parameter space. This always favours richer models because adding parameters can only increase the maximum log-likelihood; the saturated model fits perfectly by construction. One therefore seeks a criterion that trades goodness-of-fit against *model complexity*.

The two most widely used such criteria are

$$\text{AIC} = -2 \ell(\hat{\theta}; \mathbf{y}) + 2k, \quad (1)$$

$$\text{BIC} = -2 \ell(\hat{\theta}; \mathbf{y}) + k \ln n, \quad (2)$$

where k is the number of free parameters and n is the sample size. In both cases the *preferred model is the one with the smallest criterion value*.

Although the two formulae look almost identical, they arise from entirely different philosophical starting points and approximate different target quantities. The goal of this note is to derive both from first principles at a level accessible to anyone comfortable with second-order Taylor expansions and basic probability theory.

2 Preliminaries

2.1 Setup and notation

Let $\mathbf{y} = (y_1, \dots, y_n)$ be an i.i.d. sample from a true but unknown distribution with density f^* . A candidate model M parametrises a family $\{f(\cdot; \theta) : \theta \in \Theta \subset \mathbb{R}^k\}$. The log-likelihood and its maximum-likelihood estimator (MLE) are

$$\ell(\theta) = \sum_{i=1}^n \ln f(y_i; \theta), \quad \hat{\theta} = \arg \max_{\theta \in \Theta} \ell(\theta).$$

Write $\theta^* = \arg \min_{\theta \in \Theta} \text{KL}(f^* \| f(\cdot; \theta))$ for the parameter that minimises the Kullback–Leibler (KL) divergence from the truth to the model; if the model is *well-specified* then $f(\cdot; \theta^*) = f^*$.

2.2 Fisher information

The *observed* and *expected* Fisher information matrices are

$$J(\theta) = -\frac{\partial^2}{\partial \theta \partial \theta^\top} \ell(\theta), \quad \mathcal{I}(\theta) = \mathbb{E}_\theta \left[-\frac{\partial^2}{\partial \theta \partial \theta^\top} \ln f(y; \theta) \right].$$

Under standard regularity conditions $J(\hat{\theta})/n \rightarrow \mathcal{I}(\theta^*)$ almost surely as $n \rightarrow \infty$.

2.3 Asymptotic behaviour of the MLE

Under the usual regularity conditions (identifiability, smoothness of f , existence of moments),

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta^*)^{-1}). \quad (3)$$

In particular $\hat{\theta} = \theta^* + O_p(n^{-1/2})$.

3 Kullback–Leibler Divergence and its Connection to Maximum Likelihood

3.1 Definition and basic properties

The material in this section follows Wasserman [6], §§ 9.5–9.6 and 13.6; the reader is referred there for full proofs and further discussion. The original source for the KL divergence is Kullback and Leibler [1]. The classical proof of MLE consistency in the i.i.d. setting is due to Wald [2]; Kiefer and Wolfowitz [3] extend consistency to models with infinitely many incidental parameters.

Definition 3.1 (Kullback–Leibler divergence). Let f and g be probability density functions (or mass functions) on a common sample space. The *Kullback–Leibler (KL) divergence* from g to f is

$$D(f, g) = \int f(x) \log \frac{f(x)}{g(x)} dx. \quad (4)$$

For a parametric family $\{f(\cdot; \theta) : \theta \in \Theta\}$ we write $D(\theta, \psi) = D(f(\cdot; \theta), f(\cdot; \psi))$.

Remark 3.2. The notation $D(f, g)$ follows Wasserman (2004); many texts write $\text{KL}(f\|g)$. Note that $D(f, g) \neq D(g, f)$ in general, so D is *not* a distance in the formal (metric) sense. The direction matters: $D(f, g)$ measures how much information is lost when g is used to approximate f .

Proposition 3.3 (Non-negativity and definiteness). *For any densities f and g ,*

$$D(f, g) \geq 0,$$

with equality if and only if $f = g$ almost everywhere.

Proof. Apply Jensen’s inequality to the convex function $\phi(t) = -\log t$:

$$D(f, g) = \mathbb{E}_f \left[-\log \frac{g(X)}{f(X)} \right] \geq -\log \mathbb{E}_f \left[\frac{g(X)}{f(X)} \right] = -\log \int g(x) dx = -\log 1 = 0.$$

Equality in Jensen’s inequality for a strictly convex ϕ holds if and only if $g(X)/f(X)$ is constant f -a.s., which forces $f = g$ a.s. $\square$

Corollary 3.4. $D(f, f) = 0$ and, for a parametric family, $D(\theta, \theta) = 0$ for all $\theta \in \Theta$.

3.2 KL divergence in parametric families

In parametric modelling we specialise (4) by substituting density functions indexed by parameters. Let $\theta^* \in \Theta$ denote the true parameter value (or, more generally, the KL-minimising parameter as in §2). Then

$$D(\theta^*, \theta) = \int f(x; \theta^*) \log \frac{f(x; \theta^*)}{f(x; \theta)} dx = \mathbb{E}_{\theta^*} \left[\log \frac{f(X; \theta^*)}{f(X; \theta)} \right]. \quad (5)$$

By Proposition 3.3, $D(\theta^*, \theta) \geq 0$ for all θ , with $D(\theta^*, \theta^*) = 0$. Thus θ^* is the unique minimiser of $\theta \mapsto D(\theta^*, \theta)$ whenever the model is identifiable.

Definition 3.5 (Identifiability). The parametric family $\{f(\cdot; \theta) : \theta \in \Theta\}$ is *identifiable* if $\theta \neq \psi$ implies $D(\theta, \psi) > 0$, i.e. distinct parameter values correspond to distinct distributions.

We assume identifiability throughout.

3.3 The MLE as a KL minimiser

The connection between maximum likelihood and KL divergence is made precise through the *normalised log-likelihood ratio*. Define

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log \frac{f(X_i; \theta)}{f(X_i; \theta^*)}, \quad (6)$$

so that $M_n(\theta) = n^{-1}(\ell(\theta) - \ell(\theta^*))$; maximising $\ell(\theta)$ is equivalent to maximising $M_n(\theta)$ since $\ell(\theta^*)$ is a constant with respect to θ .

By the law of large numbers, for each fixed θ ,

$$M_n(\theta) \xrightarrow{P} M(\theta) := \mathbb{E}_{\theta^*} \left[\log \frac{f(X; \theta)}{f(X; \theta^*)} \right] = -D(\theta^*, \theta). \quad (7)$$

Thus the population-level objective is $M(\theta) = -D(\theta^*, \theta)$, which is maximised uniquely at $\theta = \theta^*$ (by Proposition 3.3). This is the intuition for MLE consistency: the empirical objective $M_n(\theta)$ concentrates around $-D(\theta^*, \theta)$, whose maximiser is the true θ^* .

3.4 Consistency of the MLE

Pointwise convergence (7) is not sufficient to conclude that the *maximiser* of M_n converges to the maximiser of M . We need the convergence to be *uniform*.

Theorem 3.6 (Consistency of the MLE; Wasserman 2004, Thm. 9.13). *Let θ^* denote the true parameter value. Define $M_n(\theta)$ as in (6) and $M(\theta) = -D(\theta^*, \theta)$. Suppose:*

- (i) **Uniform convergence:** $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0$.
- (ii) **Identifiability in the objective:** for every $\epsilon > 0$, $\sup_{\theta: |\theta - \theta^*| \geq \epsilon} M(\theta) < M(\theta^*)$.

Let $\hat{\theta}$ denote the MLE. Then $\hat{\theta} \xrightarrow{P} \theta^*$.

Proof sketch. Fix $\epsilon > 0$. By condition (ii) there exists $\delta > 0$ such that $M(\theta) \leq M(\theta^*) - \delta$ whenever $|\theta - \theta^*| \geq \epsilon$. By condition (i), for large enough n (with high probability), $|M_n(\theta) - M(\theta)| < \delta/2$ uniformly in θ . Hence for $|\theta - \theta^*| \geq \epsilon$:

$$M_n(\theta) \leq M(\theta) + \frac{\delta}{2} \leq M(\theta^*) - \frac{\delta}{2} \leq M_n(\theta^*).$$

The last inequality uses $M_n(\theta^*) \geq M(\theta^*) - \delta/2$. Therefore $\hat{\theta} = \arg \max M_n$ must satisfy $|\hat{\theta} - \theta^*| < \epsilon$ with high probability. Since ϵ was arbitrary, $\hat{\theta} \xrightarrow{P} \theta^*$. $\square$

Remark 3.7. Condition (i) is typically verified via a uniform law of large numbers (Glivenko–Cantelli argument), which holds under mild entropy conditions on the function class $\{\log f(\cdot; \theta) : \theta \in \Theta\}$. Condition (ii) is simply the identifiability of Definition 3.5, restated in terms of the objective function $M(\theta) = -D(\theta^*, \theta)$. The theorem shows that *MLE consistency is, at its core, a consequence of the KL divergence being minimised at the truth*.

4 Derivation of AIC

The derivation follows Akaike [7], with the risk-based presentation drawing on Konishi and Kitagawa [12], Ch. 3, and the lecture notes of Chen [5]. Burnham and Anderson [9] give an accessible account of the original argument and its practical implications.

4.1 Motivation: KL divergence and predictive quality

Akaike's (1973) key insight is to measure model quality by how well the fitted density $f(\cdot; \hat{\theta})$ approximates the true density f^* on *new* data. The natural measure is the expected KL divergence

$$\Delta(M) = \mathbb{E}_{\mathbf{y}} \left[D(f^* \| f(\cdot; \hat{\theta}(\mathbf{y}))) \right],$$

where the expectation is over the random training set $\mathbf{y}$. Expanding using (4),

$$D(f^* \| f(\cdot; \hat{\theta})) = \underbrace{\mathbb{E}_{f^*} [\ln f^*(X)]}_{\text{const w.r.t. } M} - \mathbb{E}_{f^*} [\ln f(X; \hat{\theta})].$$

Since the first term does not depend on the model, minimising $\Delta(M)$ is equivalent to maximising the *population log-likelihood* of the fitted model, evaluated on a fresh observation:

$$\bar{\ell}(\theta) := \mathbb{E}_{f^*} [\ln f(X; \theta)]. \quad (8)$$

By Proposition 3.3, $\bar{\ell}$ is maximised at θ^* ; the quantity $-\bar{\ell}(\theta^*)$ is the minimum achievable KL divergence for this model class.

4.2 Risk framework and bias

It is convenient to recast the problem in the language of statistical risk, which gives a cleaner algebraic route to the bias correction. Under the likelihood loss, the *true risk* of the MLE is

$$R(\hat{\theta}) = -n \bar{\ell}(\hat{\theta}), \quad (9)$$

the expected negative log-likelihood on fresh data, scaled by n . The *empirical risk* replaces the population expectation with the in-sample average:

$$\hat{R}(\hat{\theta}) = -\ell(\hat{\theta}) = -\sum_{i=1}^n \ln f(y_i; \hat{\theta}). \quad (10)$$

Because $\hat{\theta}$ was chosen to *maximise* ℓ on the same data used for evaluation, $\hat{R}(\hat{\theta})$ systematically underestimates $R(\hat{\theta})$. The *bias* of the empirical risk is

$$\text{bias}(M) := \mathbb{E}[\hat{R}(\hat{\theta})] - \mathbb{E}[R(\hat{\theta})] = -\mathbb{E}[\ell(\hat{\theta})] + n \mathbb{E}[\bar{\ell}(\hat{\theta})]. \quad (11)$$

The goal of AIC is to find a penalty λ such that $\hat{R}(\hat{\theta}) + \lambda$ is approximately unbiased for $R(\hat{\theta})$, which by (8) is equivalent to correcting the estimated KL divergence. Akaike showed that $\text{bias}(M) \rightarrow -k$; we derive this now.

4.3 Taylor expansion of the log-likelihood

Fix a model M with parameter $\theta \in \mathbb{R}^k$. Write

$$s(\theta) = \frac{\partial \ell}{\partial \theta}, \quad H(\theta) = \frac{\partial^2 \ell}{\partial \theta \partial \theta^\top}.$$

Expand $\ell(\theta^*)$ around the MLE $\hat{\theta}$:

$$\ell(\theta^*) = \ell(\hat{\theta}) + \underbrace{s(\hat{\theta})^\top}_{=0} (\theta^* - \hat{\theta}) + \frac{1}{2} (\theta^* - \hat{\theta})^\top H(\hat{\theta}) (\theta^* - \hat{\theta}) + O_p(n^{-1/2}). \quad (12)$$

Since $H(\hat{\theta})/n \rightarrow -\mathcal{I}(\theta^*)$ and $\hat{\theta} - \theta^* = O_p(n^{-1/2})$, the quadratic term is $O_p(1)$. Using $H(\hat{\theta}) \approx -J(\hat{\theta})$ we obtain

$$\ell(\hat{\theta}) - \ell(\theta^*) \approx \frac{1}{2} (\hat{\theta} - \theta^*)^\top J(\hat{\theta}) (\hat{\theta} - \theta^*). \quad (13)$$

4.4 Computing the bias

Theorem 4.1 (Akaike's bias formula). *Under standard regularity conditions,*

$$\mathbb{E}[R(\hat{\theta})] - \mathbb{E}[\hat{R}(\hat{\theta})] = n \mathbb{E}[\bar{\ell}(\hat{\theta})] - \mathbb{E}[\ell(\hat{\theta})] \rightarrow -k \quad \text{as } n \rightarrow \infty.$$

That is, $\text{bias}(M) \rightarrow -k$, so the empirical risk underestimates the true risk by k on average.

Proof. We compute $\mathbb{E}[\ell(\hat{\theta})] - n \mathbb{E}[\bar{\ell}(\hat{\theta})]$ and show it converges to k .

Step 1. Expand the population log-likelihood $\bar{\ell}(\hat{\theta})$ around θ^* to second order:

$$\bar{\ell}(\hat{\theta}) \approx \bar{\ell}(\theta^*) + \underbrace{\nabla \bar{\ell}(\theta^*)^\top}_{=0} (\hat{\theta} - \theta^*) + \frac{1}{2} (\hat{\theta} - \theta^*)^\top \underbrace{\nabla^2 \bar{\ell}(\theta^*)}_{=-\mathcal{I}(\theta^*)} (\hat{\theta} - \theta^*). \quad (14)$$

The gradient vanishes at θ^* because $\bar{\ell}$ is maximised there (Proposition 3.3); the Hessian is $-\mathcal{I}(\theta^*)$ by definition. Taking the expectation over $\mathbf{y}$ and multiplying by n :

$$n \mathbb{E}[\bar{\ell}(\hat{\theta})] \approx n \bar{\ell}(\theta^*) - \frac{n}{2} \mathbb{E}[(\hat{\theta} - \theta^*)^\top \mathcal{I}(\theta^*) (\hat{\theta} - \theta^*)]. \quad (15)$$

Step 2. For the expected empirical log-likelihood, write

$$\mathbb{E}[\ell(\hat{\theta})] = \mathbb{E}[\ell(\theta^*)] + \mathbb{E}[\ell(\hat{\theta}) - \ell(\theta^*)].$$

By the law of large numbers, $\mathbb{E}[\ell(\theta^*)] = n \bar{\ell}(\theta^*)$. For the second term, use the Taylor expansion (13):

$$\mathbb{E}[\ell(\hat{\theta}) - \ell(\theta^*)] \approx \frac{1}{2} \mathbb{E}[(\hat{\theta} - \theta^*)^\top J(\hat{\theta}) (\hat{\theta} - \theta^*)]. \quad (16)$$

Step 3. Since $J(\hat{\theta})/n \rightarrow \mathcal{I}(\theta^*)$ and $\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta^*)^{-1})$,

$$\frac{1}{2} \mathbb{E}[(\hat{\theta} - \theta^*)^\top J(\hat{\theta}) (\hat{\theta} - \theta^*)] \rightarrow \frac{n}{2} \mathbb{E}[(\hat{\theta} - \theta^*)^\top \mathcal{I}(\theta^*) (\hat{\theta} - \theta^*)]. \quad (17)$$

Step 4. Subtracting (15) from the expression for $\mathbb{E}[\ell(\hat{\theta})]$ using (16) and (17):

$$\mathbb{E}[\ell(\hat{\theta})] - n \mathbb{E}[\bar{\ell}(\hat{\theta})] \approx n \mathbb{E}[(\hat{\theta} - \theta^*)^\top \mathcal{I}(\theta^*) (\hat{\theta} - \theta^*)].$$

By the asymptotic normality of the MLE, $n(\hat{\theta} - \theta^*)^\top \mathcal{I}(\theta^*) (\hat{\theta} - \theta^*) \xrightarrow{d} \chi_k^2$, whose expectation is k . Hence

$$\mathbb{E}[\ell(\hat{\theta})] - n \mathbb{E}[\bar{\ell}(\hat{\theta})] \rightarrow k,$$

which is equivalent to $\text{bias}(M) \rightarrow -k$. □

4.5 The AIC formula

Theorem 4.1 says that the empirical risk $\hat{R}(\hat{\theta}) = -\ell(\hat{\theta})$ underestimates the true risk $R(\hat{\theta})$ by k on average. A bias-corrected estimator of the true risk is therefore $\hat{R}(\hat{\theta}) + k = -\ell(\hat{\theta}) + k$. Multiplying by 2 to match the conventional deviance scale:

Definition 4.2 (Akaike Information Criterion).

$$\text{AIC}(M) = -2 \ell(\hat{\theta}) + 2k.$$

A model with *smaller* AIC is preferred.

Remark 4.3. (a) The two perspectives are equivalent: minimising AIC selects the model with the smallest bias-corrected empirical risk, which by Section 3 is the same as selecting the model with the smallest estimated expected KL divergence from f^* . The risk framing gives the cleaner algebra; the KL framing gives the clearest interpretation.

(b) AIC is *not* a consistent estimator of the true best model; in large samples it tends to overfit by selecting a model that is slightly more complex than necessary. This is a direct consequence of the fixed penalty $2k$ not growing with n .

(c) A small-sample correction, $\text{AIC}_c = \text{AIC} + 2k(k+1)/(n-k-1)$, is recommended when n/k is small.

5 Derivation of BIC

The derivation follows Schwarz [8]; the Laplace approximation argument is presented in detail in Konishi and Kitagawa [12], Ch. 4, and Claeskens and Hjort [10], Ch. 3.

5.1 The Bayesian evidence

Schwarz (1978) approached model selection from a Bayesian perspective. Assign prior probabilities $P(M_j)$ to models and a prior density $\pi_j(\theta)$ to parameters within each model. Bayes' theorem gives the posterior model probability

$$P(M_j | \mathbf{y}) \propto P(M_j) p(\mathbf{y} | M_j),$$

where

$$p(\mathbf{y} | M) = \int_{\Theta} f(\mathbf{y}; \theta) \pi(\theta) d\theta \quad (18)$$

is the *marginal likelihood* or *evidence* for model M . Selecting the model with the highest posterior probability (assuming equal priors $P(M_j)$) is equivalent to maximising $p(\mathbf{y} | M)$, or equivalently minimising $-2 \ln p(\mathbf{y} | M)$.

5.2 Laplace approximation of the evidence

The integral (18) is generally intractable. The *Laplace approximation* expands the integrand around the mode of $\ln f(\mathbf{y}; \theta) \pi(\theta)$. Under mild conditions this mode approaches $\hat{\theta}$ as $n \rightarrow \infty$, because the likelihood dominates the prior.

Write $h(\theta) = \ln f(\mathbf{y}; \theta) + \ln \pi(\theta)$. Expanding to second order around $\hat{\theta}$:

$$h(\theta) \approx h(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^\top H_h(\hat{\theta})(\theta - \hat{\theta}),$$

where $H_h(\hat{\theta}) = \partial^2 h / \partial \theta \partial \theta^\top |_{\hat{\theta}} \approx -J(\hat{\theta})$ for large n , since the prior contribution is $O(1)$ while the Hessian of ℓ is $O(n)$.

Substituting into the integral and using the Gaussian integral formula:

$$\begin{aligned} p(\mathbf{y} | M) &\approx e^{h(\hat{\theta})} \int_{\mathbb{R}^k} \exp\left(\frac{1}{2}(\theta - \hat{\theta})^\top H_h(\hat{\theta})(\theta - \hat{\theta})\right) d\theta \\ &= e^{h(\hat{\theta})} \cdot (2\pi)^{k/2} | -H_h(\hat{\theta}) |^{-1/2} \\ &\approx f(\mathbf{y}; \hat{\theta}) \pi(\hat{\theta}) \cdot (2\pi)^{k/2} | J(\hat{\theta}) |^{-1/2}. \end{aligned} \quad (19)$$

5.3 The BIC approximation

Taking $-2 \ln(\cdot)$ of (19):

$$-2 \ln p(\mathbf{y} \mid M) \approx -2\ell(\hat{\theta}) - 2 \ln \pi(\hat{\theta}) - k \ln(2\pi) + \ln |J(\hat{\theta})|. \quad (20)$$

Since $J(\hat{\theta}) = \sum_{i=1}^n J_i(\hat{\theta})$ where J_i is the per-observation Hessian, we have $|J(\hat{\theta})| \approx n^k |\mathcal{I}(\hat{\theta})|$. Thus $\ln |J(\hat{\theta})| \approx k \ln n + \ln |\mathcal{I}(\hat{\theta})|$.

Retaining only the terms that grow with n (i.e., dropping all $O(1)$ terms, which cannot be consistently estimated without knowing π and $\mathcal{I}$):

Definition 5.1 (Bayesian Information Criterion).

$$\text{BIC}(M) = -2\ell(\hat{\theta}) + k \ln n.$$

A model with *smaller* BIC is preferred.

Remark 5.2. (a) The BIC is not strictly a Bayesian criterion: the $O(1)$ terms involving $\pi(\hat{\theta})$ and $|\mathcal{I}(\hat{\theta})|$ have been discarded. What remains is a rough approximation of $-2 \ln p(\mathbf{y} \mid M)$ that is easy to compute from the MLE alone.

(b) Because the penalty $k \ln n$ grows with n , BIC is a *consistent* model selection criterion under the well-specified assumption: if the true model is among the candidates, BIC selects it with probability 1 as $n \rightarrow \infty$.

(c) When $\ln n < 2$ (i.e. $n < 8$), BIC actually penalises less than AIC; in practice BIC always penalises more for $n \geq 8$.

6 Comparison of AIC and BIC

	AIC	BIC
Target quantity	Expected KL divergence / true risk	Marginal log-likelihood
Philosophy	Predictive/frequentist	Model identification/Bayesian
Penalty	$2k$	$k \ln n$
Asymptotic regime	$n/k \rightarrow \infty$	$n \rightarrow \infty, k$ fixed
Consistency	No	Yes (well-specified case)
Asymptotic efficiency	Yes	No
Large n tendency	Mild overfitting	More parsimonious

Neither criterion dominates the other uniformly. AIC targets the model with the best out-of-sample predictive performance, while BIC targets the model that generated the data. If the true model is in the candidate set and prediction is not the primary goal, BIC is preferable; if one seeks predictive accuracy or the true model is unknown, AIC is preferable.

A AIC and BIC for Generalised Linear Models

The GLM framework is due to Nelder and Wedderburn [4]; the definitive reference is McCullagh and Nelder [11]. For the use of information criteria specifically within GLMs see Burnham and Anderson [9], Ch. 7.

A.1 GLM setup

A *generalised linear model* (GLM) specifies:

1. A **random component**: $y_i \stackrel{\text{iid}}{\sim}$ from an exponential-family distribution with density

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\},$$

where θ_i is the *canonical parameter*, ϕ is the *dispersion parameter*, $b(\cdot)$ is the cumulant function, and $a_i(\phi) = \phi/w_i$ for known prior weights w_i .

2. A **systematic component**: a linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, where $\boldsymbol{\beta} \in \mathbb{R}^p$.
3. A **link function** g such that $g(\mu_i) = \eta_i$, where $\mu_i = b'(\theta_i) = \mathbb{E}[y_i]$.

The log-likelihood (up to terms not depending on $\boldsymbol{\beta}$) is

$$\ell(\boldsymbol{\beta}; \phi) = \sum_{i=1}^n \frac{w_i}{\phi} [y_i \theta_i(\boldsymbol{\beta}) - b(\theta_i(\boldsymbol{\beta}))] + \text{const.} \quad (21)$$

A.2 Deviance

The *deviance* of a fitted GLM is defined as

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2[\ell(\text{saturated}) - \ell(\hat{\boldsymbol{\beta}})] = -2[\ell(\hat{\boldsymbol{\beta}}) - \ell_{\text{sat}}], \quad (22)$$

where the *saturated model* fits one parameter per observation ($\hat{\theta}_i = \theta(y_i)$). Explicitly,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_{i=1}^n \frac{w_i}{\phi} [(y_i \tilde{\theta}_i - b(\tilde{\theta}_i)) - (y_i \hat{\theta}_i - b(\hat{\theta}_i))],$$

where $\tilde{\theta}_i$ is the canonical parameter of the saturated model ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i = \theta(\hat{\mu}_i)$.

The term $-2\ell_{\text{sat}}$ is constant across all GLMs sharing the same family and data, so comparing deviances across models of the same family is equivalent to comparing maximised log-likelihoods.

A.3 Known dispersion

When ϕ is *known* (e.g. Poisson: $\phi = 1$; binomial: $\phi = 1$), the number of free parameters is $k = p$ (the dimension of $\boldsymbol{\beta}$). Using (22):

$$-2\ell(\hat{\boldsymbol{\beta}}) = D(\mathbf{y}; \hat{\boldsymbol{\mu}}) + 2\ell_{\text{sat}} = D(\mathbf{y}; \hat{\boldsymbol{\mu}}) - 2\ell_{\text{sat}}.$$

Since $-2\ell_{\text{sat}}$ is the same for all models, we can compare models via

$$\text{AIC} = D(\mathbf{y}; \hat{\boldsymbol{\mu}}) + 2p + C, \quad (23)$$

$$\text{BIC} = D(\mathbf{y}; \hat{\boldsymbol{\mu}}) + p \ln n + C, \quad (24)$$

where $C = -2\ell_{\text{sat}}$ is a constant shared by all models under comparison. When doing relative comparisons, C cancels.

Remark A.1. In R, `AIC(fit)` and `BIC(fit)` use the full log-likelihood (including the saturated term absorbed into $c(y_i, \phi)$), so the constant C is included. Results are numerically consistent with (23)–(24).

A.4 Estimated dispersion

When ϕ is *unknown* (e.g. Gaussian, Gamma, inverse Gaussian), it must be estimated. If $\hat{\phi}$ is the MLE of ϕ , the full parameter vector is $(\boldsymbol{\beta}, \phi) \in \mathbb{R}^{p+1}$, so $k = p + 1$:

$$\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}, \hat{\phi}) + 2(p + 1), \quad (25)$$

$$\text{BIC} = -2\ell(\hat{\boldsymbol{\beta}}, \hat{\phi}) + (p + 1) \ln n. \quad (26)$$

Remark A.2. Many GLM software implementations estimate ϕ by the method-of-moments Pearson estimator rather than the MLE. When Pearson estimation is used, the log-likelihood should be evaluated at $(\hat{\boldsymbol{\beta}}, \hat{\phi}_{\text{MLE}})$, not $(\hat{\boldsymbol{\beta}}, \hat{\phi}_{\text{Pearson}})$, to maintain the theoretical justification of AIC/BIC.

A.5 Formulae for common families

Throughout let n_i denote binomial denominators where applicable, $w_i = 1$ for all i (unit weights), and $\hat{\mu}_i$ the fitted values.

Gaussian (identity link, unknown variance σ^2)

$$-2\ell(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2) = n \ln \hat{\sigma}^2 + n \ln(2\pi) + n, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2.$$

$$\text{AIC} = n \ln \hat{\sigma}^2 + 2(p + 1) + \text{const}, \quad \text{BIC} = n \ln \hat{\sigma}^2 + (p + 1) \ln n + \text{const}.$$

Poisson (log link, $\phi = 1$)

$$-2\ell(\hat{\boldsymbol{\beta}}) = 2 \sum_{i=1}^n [\hat{\mu}_i - y_i + y_i \ln(y_i / \hat{\mu}_i)] - 2 \sum_{i=1}^n [y_i - y_i \ln y_i],$$

where the second sum is constant. The deviance is $D = 2 \sum_i (y_i \ln(y_i / \hat{\mu}_i) - (y_i - \hat{\mu}_i))$.

$$\text{AIC} = D + 2p + C_{\text{Poi}}, \quad \text{BIC} = D + p \ln n + C_{\text{Poi}}.$$

Binomial (logit link, $\phi = 1$)

$$-2\ell(\hat{\boldsymbol{\beta}}) = -2 \sum_{i=1}^n [y_i \ln \hat{\mu}_i + (n_i - y_i) \ln(n_i - \hat{\mu}_i) - \ln \binom{n_i}{y_i}].$$

The deviance is $D = 2 \sum_i [y_i \ln(y_i / \hat{\mu}_i) + (n_i - y_i) \ln((n_i - y_i) / (n_i - \hat{\mu}_i))]$.

$$\text{AIC} = D + 2p + C_{\text{Bin}}, \quad \text{BIC} = D + p \ln n + C_{\text{Bin}}.$$

Gamma (log or inverse link, unknown ϕ)

Let $\nu = 1/\phi$ be the shape parameter. The log-likelihood is

$$\ell(\boldsymbol{\beta}, \nu) = n\nu \ln \nu - n \ln \Gamma(\nu) + (\nu - 1) \sum_i \ln y_i - \nu \sum_i \left(\frac{y_i}{\hat{\mu}_i} + \ln \hat{\mu}_i \right).$$

The MLE $\hat{\nu}$ satisfies $\ln \hat{\nu} - \psi(\hat{\nu}) = \overline{\ln(y/\hat{\mu})}$ (where ψ is the digamma function), solved numerically.

$$\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}, \hat{\nu}) + 2(p + 1), \quad \text{BIC} = -2\ell(\hat{\boldsymbol{\beta}}, \hat{\nu}) + (p + 1) \ln n.$$

A.6 Summary table

Family	Link	ϕ	k	Notes
Gaussian	identity	unknown (σ^2)	$p + 1$	Use MLE of σ^2
Poisson	log	1	p	
Binomial	logit	1	p	
Gamma	log/inverse	unknown ($1/\nu$)	$p + 1$	Solve for $\hat{\nu}$
Inv. Gaussian	inverse sq.	unknown	$p + 1$	
Neg. Binomial	log	unknown	$p + 1$	Shape param. estimated

B Summary

We have derived the two criteria from first principles:

- **AIC** arises from correcting the bias of the empirical risk as an estimator of the true risk. By the KL inequality, minimising true risk is equivalent to minimising expected KL divergence from the truth; the $+2k$ penalty compensates for the fact that the empirical log-likelihood overestimates the population log-likelihood by k on average, a result established via Taylor expansion and the asymptotic normality of the MLE.
- **BIC** arises from the Laplace approximation to the Bayesian marginal likelihood, retaining only the leading $O(\ln n)$ terms.

For the application to GLMs, see Appendix A: both criteria reduce to a penalty on the residual deviance, with the number of penalised parameters equal to p when ϕ is known and $p + 1$ when ϕ must be estimated by MLE.

“All models are wrong, but some are useful.” — G.E.P. Box

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